

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2121

Wayne Trace 100 kV
March 22, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2121. The Interconnection Customer proposes a 226.8 MW battery energy storage facility interconnecting at the Wayne Trace 100 kV substation in the Indiana area. The request was received on June 19, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 539 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,862,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Olander - Union-Texas 100.0 kV
Contingency: Richland - Dekalb 500.0 kV
Loading: 293.6% (Rating: 115 MVA)
DFAX: 7.5%
Pre-project loading: 278.9%
2. Facility: Ostrander - Penon Creek 100.0 kV
Contingency: Richland - Dekalb 500.0 kV
Loading: 286.8% (Rating: 115 MVA)
DFAX: 7.5%
Pre-project loading: 272.0%
3. Facility: Decatur - Penon Crk 100.0 kV
Contingency: Dekalb - Richland 500.0 kV
Loading: 271.8% (Rating: 118 MVA)
DFAX: 7.5%
Pre-project loading: 257.4%
4. Facility: Decatur Substation - S Decatur 100.0 kV
Contingency: Richland - Dekalb 500.0 kV line
Loading: 243.4% (Rating: 125 MVA)
DFAX: 7.5%
Pre-project loading: 229.8%
5. Facility: Union-Texas - Muldoon Mill 100.0 kV Line
Contingency: Dekalb - Richland Substation 500.0 kV
Loading: 238.4% (Rating: 149 MVA)
DFAX: 7.5%
Pre-project loading: 227.0%

6. Facility: Muskogee - S Decatur 100.0 kV
Contingency: Dekalb - Richland Substation 500.0 kV L
Loading: 211.6% (Rating: 136 MVA)
DFAX: 7.5%
Pre-project loading: 199.1%
7. Facility: Hassen Cassel - Darlington Rd 100.0 kV L
Contingency: Wayne Trace - Darlington Rd 100.0 kV
Loading: 175.1% (Rating: 189 MVA)
DFAX: 11.6%
Pre-project loading: 161.2%
8. Facility: Hassen Cassel - Muldoon Mill 100.0 kV
Contingency: Wayne Trace - Darlington Rd 100.0 kV
Loading: 169.1% (Rating: 204 MVA)
DFAX: 11.6%
Pre-project loading: 156.2%
9. Facility: Peavine - Segs 5 100.0 kV Line
Contingency: Segs 5 Substation - State St Substation 100.0 kV Line
Loading: 158.3% (Rating: 254 MVA)
DFAX: 29.0%
Pre-project loading: 132.3%
10. Facility: Wayne Trace - Allen 100.0 kV
Contingency: Segs 5 - Hillcrest 100.0 kV L
Loading: 150.3% (Rating: 390 MVA)
DFAX: 54.3%
Pre-project loading: 118.7%
11. Facility: State St - Segs 5 100.0 kV Line
Contingency: Dekalb - Richland 500.0 kV
Loading: 149.0% (Rating: 386 MVA)
DFAX: 18.5%
Pre-project loading: 138.2%
12. Facility: Muskogee 230 kV/100 kV Transformer Bank Unit 1
Contingency: Richland - Dekalb Substation 500.0 kV
Loading: 142.4% (Rating: 215 MVA)
DFAX: 8.9%
Pre-project loading: 133.0%
13. Facility: Muskogee 230 kV/100 kV TX T2
Contingency: Dealb - Richland 500.0 kV Line
Loading: 142.4% (Rating: 215 MVA)
DFAX: 8.9%
Pre-project loading: 133.0%
14. Facility: Muskogee 230-100 kV Transformer Bank
Contingency: Dekalb - Richland Substation 500.0 kV
Loading: 142.4% (Rating: 215 MVA)
DFAX: 8.9%
Pre-project loading: 133.0%
15. Facility: Hillcrest - Wayne Trce 100.0 kV
Contingency: Hillcrest - Muldoon Mill 100.0 kV
Loading: 140.4% (Rating: 335 MVA)
DFAX: 47.5%
Pre-project loading: 108.2%
16. Facility: Muldoon Mill - Hillcrest 100.0 kV
Contingency: Wayne Trace - Darlington Rd 100.0 kV

Loading: 133.6% (Rating: 435 MVA)
DFAX: 18.3%
Pre-project loading: 124.1%

17. Facility: State Street Substation 230/100 kV Transformer Bank T1

Contingency: Dekalb - Richland 500.0 kV
Loading: 129.5% (Rating: 443 MVA)
DFAX: 7.4%
Pre-project loading: 125.8%

18. Facility: State Street 230/100kV Auto #2

Contingency: Richland Substation - Dekalb 500.0 kV line
Loading: 129.5% (Rating: 443 MVA)
DFAX: 7.4%
Pre-project loading: 125.8%

19. Facility: State St 230/100kV Transformer Unit 3

Contingency: Dekalb - Richland 500.0 kV
Loading: 129.5% (Rating: 443 MVA)
DFAX: 7.4%
Pre-project loading: 125.8%

20. Facility: State St 230 kV/100 kV TX Unit 4

Contingency: Richland Substation - Dekalb 500.0 kV
Loading: 129.5% (Rating: 443 MVA)
DFAX: 7.4%
Pre-project loading: 125.8%

21. Facility: Stte St - Peaine 100.0 kV Line 1

Contingency: Richland - Dekalb 500.0 kV L
Loading: 126.8% (Rating: 332 MVA)
DFAX: 6.9%
Pre-project loading: 122.0%

22. Facility: State Street - Peavine 100.0 kV Cir. 2

Contingency: Richland - Dekalb 500.0 kV
Loading: 126.8% (Rating: 332 MVA)
DFAX: 6.9%
Pre-project loading: 122.0%

23. Facility: Aep Pettit Avenue - Allen 100.0 kV

Contingency: Segs 5 - Hillcrest 100.0 kV
Loading: 123.1% (Rating: 450 MVA)
DFAX: 54.3%
Pre-project loading: 95.8%

24. Facility: Darlington Rd - Wayne Trace 100.0 kV Line

Contingency: Richland - Dekalb 500.0 kV
Loading: 109.3% (Rating: 702 MVA)
DFAX: 27.1%
Pre-project loading: 100.5%

25. Facility: Hillcrest - Segs 5 100.0 kV

Contingency: Dekalb - Richland 500.0 kV
Loading: 107.4% (Rating: 840 MVA)
DFAX: 46.1%
Pre-project loading: 94.9%

26. Facility: Breed - Sidney A Murray Jr Hydroelectric 230.0 kV L

Contingency: Richland - Dekalb 500.0 kV
Loading: 100.5% (Rating: 2,568 MVA)

DFAX: 10.2%
Pre-project loading: 99.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ostander - Union-Texas 100.0 kV	Rebuild line with higher-capacity conductor	\$3,817,000	72% (\$2,730,000)
Ostrander - Peno Crk 100.0 kV	Rebuild line with higher-capacity conductor	\$2,977,000	72% (\$2,129,000)
Decatur - Peno Creek 100.0 kV	Rebuild line with higher-capacity conductor	\$8,653,000	72% (\$6,189,000)
Decatur Substation - S Decatur 100.0 kV	Rebuild line with higher-capacity conductor	\$5,014,000	72% (\$3,586,000)
Union-Texas - Muldoon Mill 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,581,000	72% (\$1,846,000)
Musogee - S Decatur 100.0 kV	Rebuild line with higher-capacity conductor	\$10,564,000	72% (\$7,556,000)
Hassen Cassel - Darlington Rd 100.0 kV L	Rebuild line with higher-capacity conductor	\$3,038,000	62% (\$1,884,000)
Hassen Cassel - Muldoon Mill 100.0 kV	Rebuild line with higher-capacity conductor	\$2,635,000	62% (\$1,634,000)
Segs 5 - Peavine 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,816,000	19% (\$345,000)
Wayne Trace - Allen 100.0 kV	Rebuild line with higher-capacity conductor	\$6,746,000	31% (\$2,118,000)
State St - Segs 5 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,317,000	13% (\$307,000)
Muskogee 230-100 kV TX	Replace transformer with higher-rated unit	\$7,316,000	72% (\$5,232,000)
Muskogee 230 kV/100 kV TX T2	Replace transformer with higher-rated unit	\$14,362,000	72% (\$10,272,000)
Muskogee 230-100 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,837,000	72% (\$4,175,000)
Hillcrest - Wayne Trce 100.0 kV	Rebuild line with higher-capacity conductor	\$1,946,000	30% (\$589,000)
Muldoon Mill - Hillcrest Substation 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,657,000	13% (\$868,000)
State St 230/100kV TX Bank 1	Replace transformer with higher-rated unit	\$8,499,000	6% (\$518,000)
State Street 230/100kV Auto #2	Replace transformer with higher-rated unit	\$10,490,000	6% (\$639,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
State St 230/100kV Transformer Unit 3	Replace transformer with higher-rated unit	\$6,149,000	6% (\$375,000)
State St 230/100 kV Autotransformer	Replace transformer with higher-rated unit	\$11,593,000	6% (\$707,000)
Stte St - Peaine 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$2,029,000	7% (\$137,000)
State Street - Peavine 100.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,394,000	7% (\$162,000)
Aep Pettit Avenue - Allen 100.0 kV	Rebuild line with higher-capacity conductor	\$7,850,000	31% (\$2,446,000)
Darlington Rd - Wayne Trace 100.0 kV Line	Reconductor line	\$319,000	12% (\$38,000)
Hillcrest - Segs 5 100.0 kV	Reconductor line	\$1,290,000	15% (\$195,000)
Sidney A Murray Jr Hydroelectric - Breed 230.0 kV	Reconductor line	\$12,721,000	2% (\$185,000)
	Total allocated to project		\$56,862,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.